

Supplementary Material: Fixed Local Pallet Refinement

Internal draft for author review

September 2026

1 Status and reading guide

This supplement preserves the distinction between completed development evidence and independent confirmation. The machine-readable status is in `FINAL_STATUS.json` under the submission experiment directory. All three matched-budget fits were completed; the learning curves are diagnostic, not a best-checkpoint selector. Independent real-session confirmation is not yet available. No method’s incomplete result is represented by another model or a zero-valued measurement.

2 Full development results

Original-pixel medians and P90 are pooled over each model’s supervised matched points. Frame mean is the mean of matched per-frame errors, not the pooled median. Gross20 and PCK are percentages; PCK retains the full 2,818-point ground-truth denominator. Raw prior rows, when available, precede no claim that the selected capped wrapper is the original standalone method.

Model	Med	P90	Mean	Gross20	PCK5	PCK10	PCK20	N
R0	6.616	38.670	21.673	17.20	37.19	63.73	80.98	2756
P1	5.851	38.025	20.930	15.57	42.26	67.64	82.58	2756
P2	5.943	37.728	21.010	15.89	42.16	67.07	82.26	2756
P3	5.921	37.546	20.949	15.42	41.66	67.32	82.72	2756
L1	6.052	37.204	21.145	16.07	41.23	66.57	82.08	2756
L2	6.037	37.302	21.085	15.93	41.06	66.93	82.22	2756
L3	6.121	37.453	21.164	16.00	41.02	66.86	82.15	2756
D1	6.366	37.479	21.411	16.11	39.14	64.66	82.04	2756
D2	6.457	37.810	21.456	16.44	38.75	64.62	81.72	2756
D3	6.467	37.410	21.435	16.22	38.96	64.44	81.94	2756
PRIOR1	5.569	37.218	20.784	15.49	44.78	68.31	82.65	2756
PRIOR1_raw	5.569	36.776	20.655	15.42	44.78	68.28	82.72	2756
PRIOR2	5.541	37.373	20.738	15.64	44.75	68.84	82.51	2756
PRIOR2_raw	5.541	37.295	20.594	15.53	44.78	68.88	82.61	2756
PRIOR3	5.596	37.089	20.830	15.71	43.90	68.06	82.43	2756
PRIOR3_raw	5.599	36.721	20.700	15.60	43.90	68.10	82.54	2756

3 Geometry-reference results

Rotation and yaw are degrees; translation is centimeters. These are not independent physical 6D measurements. Pose coverage is separate from box matching: the canonical prediction-only selector may return a pose from a top detection that does not meet the 2D matching gate. No new symmetry minimization was introduced to excuse near-square axis confusion.

Model	Rot(deg)	Yaw(deg)	Trans(cm)	IoU3D	ADD AUC	Coverage
R0	2.262	1.231	7.897	0.603	0.428	1.000
P1	2.089	1.074	6.865	0.642	0.454	1.000
P2	2.106	1.136	6.917	0.637	0.456	1.000
P3	2.073	1.156	7.678	0.637	0.454	1.000

Model	Rot(deg)	Yaw(deg)	Trans(cm)	IoU3D	ADD AUC	Coverage
L1	2.110	1.115	7.609	0.637	0.450	1.000
L2	2.103	1.122	7.364	0.636	0.452	1.000
L3	2.136	1.132	7.672	0.625	0.445	1.000
D1	2.201	1.192	7.570	0.621	0.441	1.000
D2	2.181	1.182	7.540	0.619	0.442	1.000
D3	2.188	1.192	7.682	0.616	0.440	1.000
PRIOR1	2.018	1.100	7.185	0.614	0.455	1.000
PRIOR1_raw	2.018	1.100	7.185	0.616	0.456	1.000
PRIOR2	2.004	1.102	6.675	0.628	0.467	1.000
PRIOR2_raw	2.010	1.105	6.844	0.628	0.464	1.000
PRIOR3	2.132	1.190	6.825	0.619	0.460	1.000
PRIOR3_raw	2.132	1.190	6.825	0.619	0.460	1.000

4 Registered geometry and camera inputs

Object	Long(m)	Short(m)	Height(m)	Frames
Plastic	1.30	1.10	0.11	194
Wood	0.80	0.59	0.14	125

Width	Height	f_x	f_y	c_x	c_y	Frames
640	480	605.906	605.970	317.596	256.292	28
640	480	608.298	607.846	326.401	239.547	146
640	480	614.184	614.313	329.280	234.529	100
1280	720	908.860	908.955	636.394	384.438	45

Intrinsics above are annotation records in pixels, not newly measured calibration accuracy. The canonical solver passes no distortion coefficients. The records inspected here provide resolution and intrinsics, not an independently verified lens/calibration uncertainty report. Object dimensions are the unchanged registered geometry; the evaluation contract does not claim new physical inspection. The declared long-axis equivalence allows a half turn, but not a quarter turn. The wood object is not established as visually symmetric under the half turn. These benchmark conventions do not certify interchangeable physical fork-entry geometry.

5 Complete timing summaries

The expanded panel was measured in the same session for all comparators.

All values are milliseconds. The same-session expanded panel is used after the prior evaluation is complete; otherwise only the explicitly historical timing is available. Every measured repeat is retained. Model and image loading are excluded, but actual RGB preprocessing, detector forward, feature sampling/refinement, restoration and canonical PnP are included in their named intervals. Allocated-memory figures do not include unrelated desktop processes.

Model	2D med	mean	P90	Total med	mean	P90	Add2D med
R0	9.733	10.017	11.803	11.217	11.522	13.298	0.000
P1	13.661	14.031	15.730	15.352	15.496	17.096	4.056
P2	13.622	13.996	15.886	15.280	15.465	17.332	3.802
P3	13.784	14.096	15.984	15.550	15.596	17.355	4.077
L1	18.226	18.106	20.105	19.736	19.598	21.496	7.975
L2	18.150	18.163	20.161	19.858	19.670	21.572	8.115
L3	18.178	18.194	20.097	19.734	19.685	21.457	8.194
D1	13.231	13.509	15.482	14.800	15.022	16.941	3.343
D2	12.979	13.464	15.558	14.800	14.993	16.948	3.284
D3	13.226	13.444	15.365	14.773	14.927	16.832	3.211

Model	2D med	mean	P90	Total med	mean	P90	Add2D med
PRIOR1	27.537	27.380	28.439	29.020	28.926	29.890	17.466
PRIOR2	27.844	27.824	28.737	29.320	29.308	30.179	17.937
PRIOR3	27.844	27.653	28.594	29.297	29.123	29.994	17.918

The first complete balanced panel supplies every latency sample. An earlier incomplete attempt failed an overly strict bit-exact replay check and is retained separately, not selected or rejected by speed. Fixed-input repeated GPU forwards located the prior’s first numerical differences in its first upsampling stage; the stem and ResNet blocks were identical in that diagnostic. The frozen nondeterministic cuDNN settings and all model weights remain unchanged. Corrected prior coordinates are checked in crop space using the existing network validation’s absolute component, 3×10^{-4} pixels, with zero relative allowance. Detector contracts and center8 remain bit-exact. Historical coordinates reproduce canonical pose metrics; current prior coordinates must preserve hypothesis and coverage. The final DEV accuracy tables still use the original frozen predictions.

After all balanced timing samples passed, the model-release guard prevented reuse of the same process for isolated memory measurements. The saved latency samples were retained unchanged; allocation was measured in a new process for each model. A further untimed coordinate replay audit belongs to that memory phase, not to the earlier timing samples. Per-sample timed numerical bounds passed, but individual drift magnitudes and OpenCV/inter-operation thread observations were not saved before the guard stopped the process. Those missing observations are explicitly distinguished from the separately recorded memory-phase settings. Recovery records and both original attempts are retained; no fastest session is chosen.

6 Controlled training and convergence limits

Parameter, memory and cache accounting

Model	Learned	Total	Add/R0(%)	Inf.MiB	Train MiB	s/update
R0	0	2615142	0.00	190.01	NM	NM
P1	18962	2634104	0.73	190.09	NM	0.0590
D1	19450	2634592	0.74	190.09	NM	0.0662
L1	19810	2634952	0.76	190.09	NM	NM
PRIOR1	68661641	71276783	2625.54	454.84	7779.02	0.4163

The shared R0 has 2,615,142 parameters. The recorded first cache pass processed 60,000 records in 1668.83 seconds; bound cache arrays occupy 68.677 GiB. Head fit time excludes that extraction. NM indicates an unavailable historical allocation measurement, not zero cost. R0 is not newly trained. CPU: 11th Gen Intel(R) Core(TM) i9-11900 @ 2.50GHz. Timed code fixes Torch intra-operation threads to 4. Inter-operation and OpenCV thread observations were not durably captured in the timed phase. In the separate fresh-process memory phase, observed settings were Torch intra-operation 4, inter-operation 8, OpenCV 1 (R0 process; all models are in the runtime JSON). These are not retrospectively labeled timed-phase observations. P and D share source order, feature evidence and historical optimizer/exposure settings, but their loss and output geometry differ. D’s small fixed-probe decrease is neither a convergence guarantee nor evidence of fundamental regression inferiority. The prior uses ImageNet TF-Slim ResNet152 initialization and the official loss/Adam semantics adapted to nine pallet points. Its sample order is exactly the existing P order and its nominal batch is sixteen. Source train/cal probes at step0 and every500 updates are diagnostic only. No checkpoint or wrapper is selected on DEV.

7 Operator and weight provenance

The official PoseFix source commit is 5556364bb0f43b0743a5fcd820de48f34b3d4360. Its MIT notice and the TensorFlow authors’ Apache-2.0 attribution are retained. The official TF1 CPU reference runs in an isolated Python3.7 environment, and the PyTorch port uses the existing YOLO runtime without replacing that environment’s Torch, CUDA, or NumPy.

The reference initializes new weights using the official graph and transfers ImageNet RGB/backbone variables. Added pose channels remain at the official seeded initialization. The port maps convolution HWIO to OIHW,

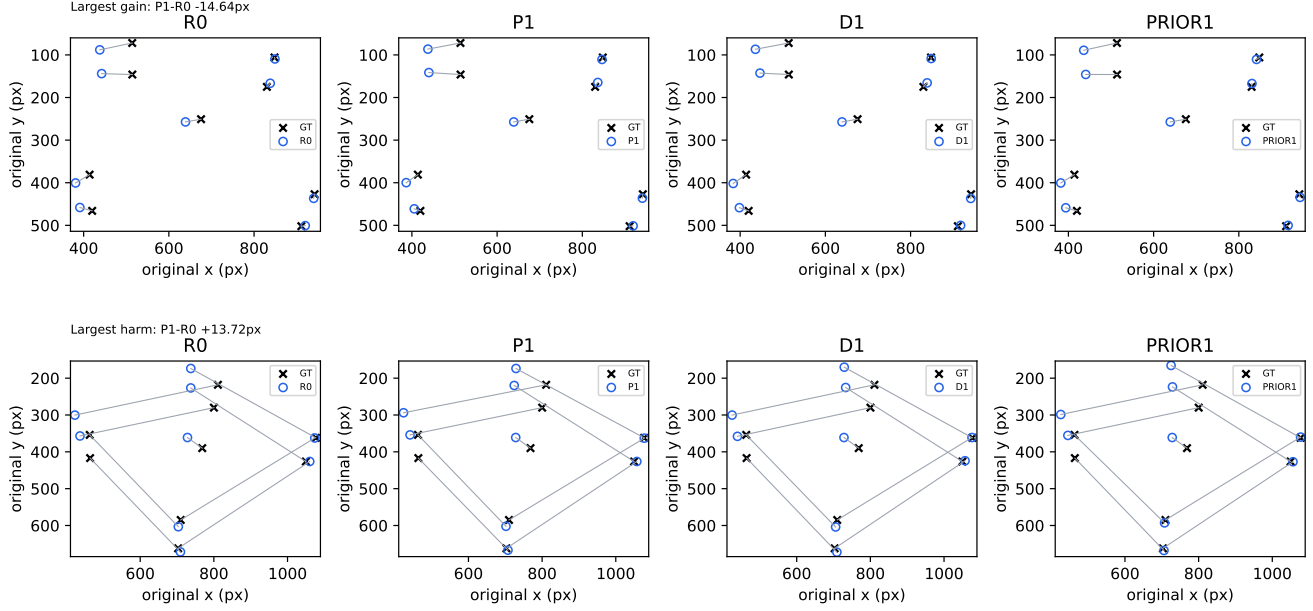


Figure 1: Actual improving and harming DEV cases: the lowest and highest per-frame median difference P1–R0, with lexical frame-ID ties. Same-frame GT and every plotted model use unchanged original coordinates and shared limits. Membership and raw sources are in `FIGURE_MANIFEST.json`. Images are not republished without verified image rights; these coordinate-only panels do not establish physical boundary correctness or independent generalization.

transpose convolution HWOI to IOHW, and every BN affine/moving variable. Missing and unexpected keys are explicitly checked. Full eval-coordinate and stage-feature comparisons use the actual source images and the same weights; scalar/loss-gradient and moving-statistics comparisons separately test training operators.

The graph’s nominal BN epsilon is not its effective value: the TF1 fused wrapper enforces 1.001×10^{-5} . The torch port matches this executed behavior. The Gaussian amplitude is 255 exactly once. Expectation coordinates are zero-based and scaled by 4 at the locked resolution. Bilinear target neighbors outside the grid are excluded and remaining mass normalized; nonfinite/outside-crop targets are masked in training, without altering evaluation membership. Raw RGB is explicitly the requested adaptation from the inspected BGR-preserving source loader.

The full batch16 resource probe and actual parameter update are recorded separately from the three main fits. Main initialization and optimizer state do not reuse that smoke update. Checkpoints contain optimizer, Python/NumPy/CPU/CUDA RNG, step and exact order-prefix hash. The loader has no hidden random augmentation state; pending RGB reads can be reconstructed deterministically from the next recorded row.

8 Research history and interpretation

Prior Hough/line, active-adaptation, replay and robustness investigations motivated subsequent decisions, but do not become additional unseen validation sets. Old code, checkpoints, metrics and verdicts remain unchanged. This selection history means DEV is reused development evidence. The fixed model is not redesigned in response to an unfavorable prior comparison.

The baseline checkpoint is not purely synthetic-from-scratch: the recorded initialization checkpoint identifies COCO-pose training, despite a historical output directory containing the word scratch. P’s additional training uses synthetic labels only. ImageNet supervision for the added prior is separately disclosed. These distinctions prevent an overbroad claim that no real annotation contributed anywhere.

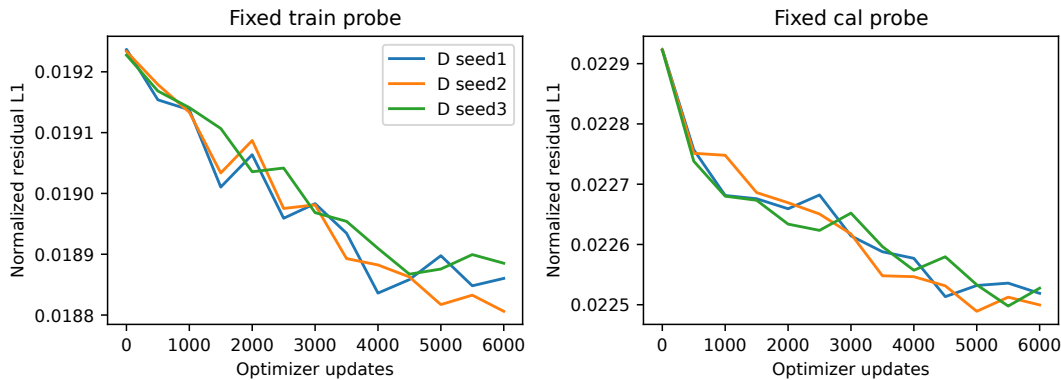


Figure 2: Already-completed D fixed probes. Their limited change is disclosed alongside the comparison, not suppressed. Loss magnitudes are not compared with P’s different objective.

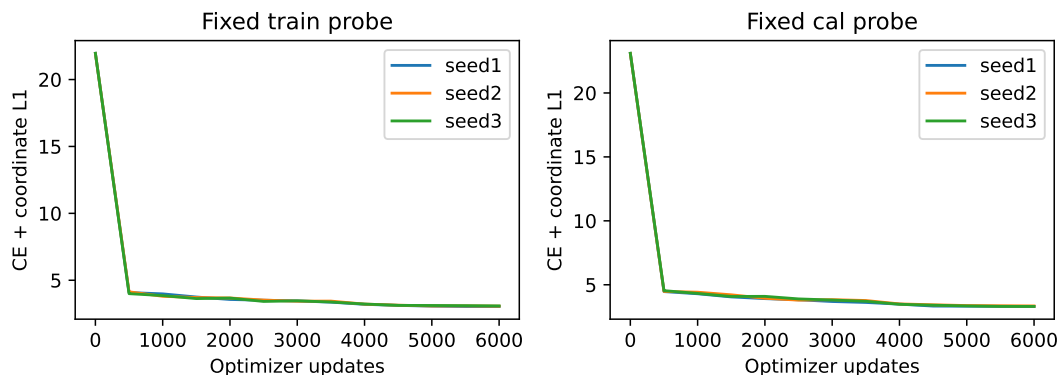


Figure 3: Prior fixed train/cal probes from the completed bounded fits. These curves did not select a checkpoint or extend training.

9 Independent data and annotation handoff

The actual collection and schema are in `HUMAN_INPUTS_REQUIRED.txt` and `CONFIRMATION_SCHEMA.json`. The starting plan is 12 separate sessions with 15 prespecified frames each; at least 30 frames receive two independent prediction-blinded annotations and blind adjudication. The count is not a power guarantee. Preserve original videos, timestamps, capture groups, camera/mount, calibration, actual object dimensions/identity and prior-use declarations. Changed image encoding or filename does not prove independence from an old scene.

Unreviewed provenance and an unfrozen comparator panel block confirmation only, not the completed development experiment. New-frame reference geometry must be bound to its camera/dimensions/index convention before a 6D claim. Human annotation, author identities, funding and consent cannot be fabricated by the CLI. These obligations are separate from compilation and code checks.

10 Reproduction and access

From the repository root, use the recorded environment and `run.py status`, `run.py all`, or `run.py resume` under `scripts/research/pallet_sensors_submission_v1/`. Exact shell commands, downloads, locks and stage receipts are in `REPRODUCE.md`. Completion receipts are checked against actual input/output hashes, not accepted solely on a status string. Main uses no branch rewrite, force push or deletion of unrelated user changes.

Public materials contain code, small numerical results, plots and manuscript sources. Raw training/evaluation imagery, video, feature caches, model weights and the isolated environment remain excluded from public Git. Access restrictions and licenses must be reviewed before any separate sharing. No unverified speedup, annotation-cost reduction, or public-data right is inferred.